

Soham Gunturu

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Education

Georgia Institute of Technology

Bachelor of Science in Computer Science (AI and Systems Architecture Threads)

Atlanta, GA

Aug 2024 – May 2028

- **GPA:** 3.9 / 4.0
- **Relevant Coursework:** Operating Systems, Advanced Computer Architecture, Compilers & Interpreters, Machine Learning, Data Structures & Algorithms, Systems & Networks, Probability & Statistics, Discrete Math
- **Activities:** Trading Club (Exec), AI@GT, Hexlabs, GROWER Lab, Ganesh Lab, Taal Tadka A Cappella (President)

Experience

Databricks

Software Engineer Intern

May 2026 – Aug 2026

Bellevue, WA

- Built persistence capabilities for different data manipulations in the datagrid, elevating customer satisfaction by 17%.
- Built a data-prep workflow in TypeScript/MobX, shipping 12 grid actions as reusable operators in Lakeflow Designer.
- Engineered PySpark-backed data operators for duplicate removal and null handling, improving product reach by 23%.

Millennium

Meet Millennium Participant

Apr 2026

Miami, FL

- 1 of 25 students selected out of thousands of engineering applicants for Millennium's Miami technology track.
- Explored trading systems & strategies through lectures and an algorithmic trading game, winning 1st out of 7 teams.

Microsoft

Product Fellow

Sep 2025 – Jan 2026

Atlanta, GA

- Developed a full-stack agentic incident management system, estimated to save \$500K annually in developer effort.
- Trained a TF-IDF + Logistic Regression pipeline to route incidents to the correct team with 85%+ accuracy.
- Contextualized tickets by surfacing similar historical cases and relevant resources, reducing triage time by 40%.

Amazon

Software Development Engineer Intern

May 2025 – Aug 2025

Tempe, AZ

- Built internal tooling to detect and resolve conflicting custom product offers, cutting error diagnosis time by 20%.
- Raised Buyer Preview usage by 22% by building a full-stack feature for sellers to download a sample fulfillment file.
- Sped up deployment pipeline by 30% through optimization of Seller Central frontend and backend test suites.

Vijay Ganesh Lab

Undergraduate Researcher

Apr 2025 – Present

Atlanta, GA

- Integrating SAT Solver mechanisms with LLMs to enhance model reasoning and learning capabilities by 20%.
- Building a Pydantic AI multi-agent system to apply Conflict-Driven Clause Learning (CDCL) to reasoning models.

Projects

Real-time AR assistant | *Swift, ARKit, YOLOv8, Python, FastAPI, PostgreSQL, AVFoundation*

Mar 2026

- Building an iOS AR assistant which constructs custom task plans with directional awareness and AR overlays.
- Creating an on-device CV pipeline with sub-200ms latency, enabling real-time object detection and annotation.
- Designing a FastAPI backend with 12 REST routers covering vision, step generation, chat history, JWT auth, etc.

xv6 Kernel | *C, Assembly*

Jan 2026

- Implemented copy-on-write forking and lazy zero-page allocation in xv6, reducing fork() clock cycles by 99%.
- Extended xv6 with priority-aware schedulers, clone()-based threading, and userspace synchronization primitives.
- Built an Intel e1000 NIC driver with PCI enumeration, DMA ring buffers, and sub-ms Ethernet frame syscalls.

HFT Simulator for RL agents | *C++, Python, PyBind, PyTorch, Gymnasium, Numpy, Stable-Baseline3*

May 2025

- Built a high-frequency market simulator in C++, processing 10,000+ orders/sec with realistic microstructure.
- Designed an RL environment supporting 18-D state spaces, enabling strategy agent benchmarking with tracked PnL.
- Achieved 27% faster simulation throughput over pure Python by implementing order book engine in C++.

Volatility Arbitrage Engine | *Python, Matplotlib, Pandas, Scikit-Learn, Numpy, Jupyter*

Apr 2025

- Created an engine to find vol arbitrage between an ETF and constituents using PCA decomp and options pricing.
- Back-tested the signal on historical options data across 900+ ETFs, achieving 32% average annualized returns.

Skills and Interests

Languages: Python, Java, C, C++, JavaScript, TypeScript, SQL, HTML, CSS, Assembly, Swift, R

Frameworks: Node.js, React.js, Express.js, FastAPI, Pydantic AI, LangChain, React Native, Flask, Spring Boot, Dagger

Software: Git, PyTorch, Transformers, Pandas, Scikit-Learn, PostgreSQL, Firebase, AWS, MongoDB, Docker, Kubernetes

Awards: Presidential Undergrad Research Award, Ohio Governor's Scholar, Hack OHI/O Winner, Presidential Gold Award

Interests: Chess, Poker, Music, Travel, Basketball, Climbing, Chess